

CHAPTER C27

WIND LOADS ON BUILDINGS: MAIN WIND FORCE RESISTING SYSTEM (DIRECTIONAL PROCEDURE)

The Directional Procedure is the former “buildings of all heights” provision in Method 2 of ASCE 7-05 for the main wind force resisting system (MWFRS). A simplified method based on this Directional Procedure is provided for buildings up to 160 ft (49 m) in height. The Directional Procedure is considered the traditional approach in that the pressure coefficients reflect the actual loading on each surface of the building as a function of wind direction, namely, winds perpendicular or parallel to the ridge line.

C27.1 SCOPE

C27.1.5 Minimum Design Wind Loads. This section specifies a minimum wind load to be applied horizontally on the entire vertical projection of the building, as shown in Fig. C27.1-1. This load case is to be applied as a separate load case in addition to the normal load cases specified in other portions of this chapter.

PART 1: ENCLOSED, PARTIALLY ENCLOSED, AND OPEN BUILDINGS OF ALL HEIGHTS

C27.3 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

C27.3.1 Enclosed and Partially Enclosed Rigid and Flexible Buildings. In Eqs. (27.3-1) and (27.3-2), a velocity pressure term, q_i , appears that is defined as the “velocity pressure for internal pressure determination.” The positive internal pressure is dictated by the positive exterior pressure on the windward face at the point where there is an opening. The positive exterior pressure at the opening is governed by the value of q at the level of the opening, not q_h . For positive internal pressure evaluation, q_i may conservatively be evaluated at height h ($q_i = q_h$). For low buildings, this evaluation does not make much difference, but for the example of a 300-ft- (91.4-m)-tall building in Exposure B with a highest opening at 60 ft (18.2 m), the difference between q_{300} and q_{60} represents a 59% increase in internal pressure. This difference is unrealistic and represents an unnecessary degree of conservatism. Accordingly, $q_i = q_z$ for positive internal pressure evaluation in partially enclosed buildings where height z is defined as the level of the highest opening in the building that could affect the positive internal pressure. For buildings sited in wind-borne debris regions, with glazing that is not impact-resistant or protected with an impact-protective system, q_i should be treated on the assumption that there is an opening.

Fig. 27.3-1. The pressure coefficients for MWFRSs are separated into two categories:

1. Directional Procedure for buildings of all heights (Fig. 27.3-1) as specified in Chapter 27 for buildings that meet the requirements specified therein.
2. Envelope Procedure for low-rise buildings that have a height less than or equal to 60 ft (18.3 m) (Fig. 28.3-1) as specified in Chapter 28 for buildings that meet the requirements specified therein.

In generating these coefficients, two distinctly different approaches were used. For the pressure coefficients given in Fig. 27.3-1, the more traditional approach was followed, and the pressure coefficients reflect the actual loading on each surface of the building as a function of wind direction, namely, winds perpendicular or parallel to the ridge line.

Observations in wind tunnel tests show that areas of very low negative pressure and even slightly positive pressure can occur in all roof structures, particularly as the distance from the windward edge increases and the wind streams reattach to the surface. These pressures can occur even for relatively flat or low-slope roof structures. Experience and judgment from wind tunnel studies have been used to specify either zero or slightly negative pressures (-0.18) depending on the negative pressure coefficient. These values require the designer to consider a zero or slightly positive net wind pressure in the load combinations of Chapter 2.

Fig. 27.3-2. Frame loads on dome roofs are adapted from the Eurocode (1995). The loads are based on data obtained in a modeled atmospheric boundary-layer flow that does not fully comply with requirements for wind tunnel testing specified in this standard (Blessman 1971). Loads for three domes ($h_D/D = 0.5$, $f/D = 0.5$), ($h_D/D = 0$, $f/D = 0.5$), and ($h_D/D = 0$, $f/D = 0.33$) are roughly consistent with data of Taylor (1991), who used an atmospheric boundary layer as required in this standard. Two load cases are defined, one of which has a linear variation of pressure from A to B as in the Eurocode (1995), and one in which the pressure at A is held constant from 0° to 25° ; these two cases are based on comparison of the Eurocode provisions with Taylor (1991). Case A (the Eurocode calculation) is necessary in many cases to define maximum uplift. Case B is necessary to properly define positive pressures for some cases, which cannot be isolated with current information and which result in maximum base shear. For domes larger than 200 ft (61 m) in diameter, the designer should consider use of wind tunnel testing. Resonant response is not considered in these provisions. Wind tunnel testing should be used to consider resonant response. Local bending moments in the dome shell may be larger than predicted by this method because of the difference between instantaneous local pressure distributions and those predicted by Fig. 27.3-2.

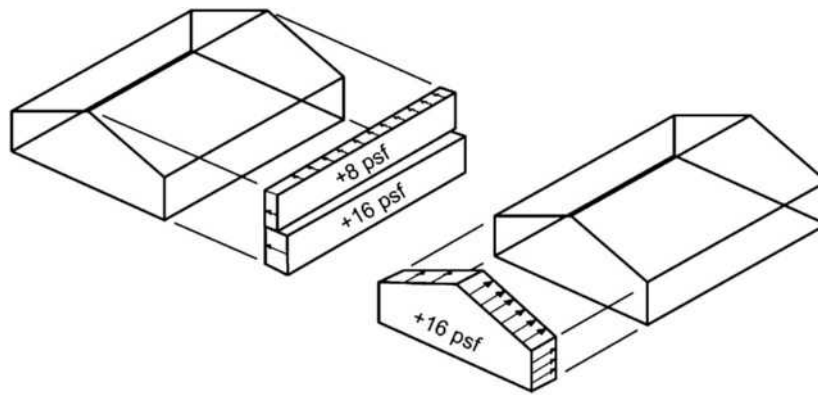


FIGURE C27.1-1. Application of Minimum Wind Load

Note: 1.0 lb/ft² = 0.0479 kN/m²

If the dome is supported on vertical walls directly below, it is appropriate to consider the walls as a “chimney” using Fig. 29.4-1.

Fig. 27.3-3. The pressure and force coefficient values in these tables were taken from ANSI A58.1-1972 (1972). Additional information was added in ANSI A58.1-1982 (1982), which included use of a 1.2 multiplier for component and cladding (C&C) external pressure coefficients, other than for the perimeter areas. That multiplier was changed from 1.2 to 0.87 in ASCE 7-95 (1998), however, no substantiation was provided for the change. The multiplier is changed back to 1.2 in ASCE 7-16. The coefficients specified in these tables are based on wind tunnel tests conducted under conditions of uniform flow and low turbulence, and their validity in turbulent boundary-layer flows has yet to be completely established. Additional pressure coefficients for conditions not specified herein may be found in SIA (1956) and ASCE (1961).

C27.3.2 Open Buildings with Monoslope, Pitched, or Troughed Free Roofs. Figs. 27.3-4 through 27.3-6 and 30.7-1 through 30.7-3 are presented for wind loads on MWFRSs and C&C of open buildings with roofs as shown, respectively. This work is based on the Australian standard AS1170.2-2002, Part 2: Wind Actions, with modifications to the MWFRS pressure coefficients based on recent studies (Altman 2001; Uematsu and Stathopoulos 2003).

Two load cases, A and B, are given in Figs. 27.3-4 through 27.3-6. These pressure distributions provide loads that envelop the results from detailed wind tunnel measurements of simultaneous normal forces and moments. Application of both load cases is required to envelop the combinations of maximum normal forces and moments that are appropriate for the particular roof shape and blockage configuration.

The roof wind loading on open building roofs is highly dependent upon whether goods or materials are stored under the roof and restrict the wind flow. Restricting the flow can introduce substantial upward-acting pressures on the bottom surface of the roof, thus increasing the resultant uplift load on the roof. Figs. 27.3-4 through 27.3-6 and 30.7-1 through 30.7-3 offer the designer two options. Option 1 (clear wind flow) implies that little (less than 50%) or no portion of the cross section below the roof is blocked. Option 2 (obstructed wind flow) implies that a significant portion (more than 75% is typically referenced in the literature) of the cross section is blocked by goods or materials below the roof. Clearly, values would change from one set of coefficients to the other following some sort of

smooth, but as yet unknown, relationship. In developing the provisions included in this standard, the 50% blockage value was selected for Option 1, with the expectation that it represents a somewhat conservative transition. If the designer is not clear about usage of the space below the roof or if the usage could change to restrict free air flow, then design loads for both options should be used.

See Section 28.3.5 for explanation of the horizontal wind loads on open buildings with transverse frames and pitched roofs that act in combination with the loads calculated in Section 27.3.3.

C27.3.5 Design Wind Load Cases. Wind tunnel research (Isyumov 1983; Boggs et al. 2000; Isyumov and Case 2000; Xie and Irwin 2000) has shown that torsional load is caused by nonuniform pressure on the different faces of the building from wind flow around the building, interference effects of nearby buildings and terrain, and dynamic effects on more flexible buildings. Load Cases 2 and 4 in Fig. 27.3-8 specifies the torsional loading to 15% eccentricity under 75% of the maximum wind shear for Load Case 2. Although this is more in line with wind tunnel experience on square and rectangular buildings with aspect ratios up to about 2.5, it may not cover all cases, even for symmetric and common building shapes where larger torsions have been observed. For example, wind tunnel studies often show an eccentricity of 5% or more under full (not reduced) base shear. The designer may wish to apply this level of eccentricity at full wind loading for certain more-critical buildings, even though it is not required by the standard. The present more-moderate torsional load requirements can, in part, be justified by the fact that the design wind forces tend to be upper bound for most common building shapes.

In buildings with some structural systems, more severe loading can occur when the resultant wind load acts diagonally to the building. To account for this effect and the fact that many buildings exhibit maximum response in the across-wind direction (the standard currently has no analytical procedure for this case), a structural system should be capable of resisting 75% of the design wind load applied simultaneously along each principal axis, as required by Case 3 in Fig. 27.3-8.

For flexible buildings, dynamic effects can increase torsional loading. Additional torsional loading can occur because of eccentricity between the elastic shear center and the center of mass at each level of the building. Eq. (27.3-4) accounts for this effect.

It is important to note that significant torsion can occur on low-rise buildings also (Isyumov and Case 2000), and therefore, the wind loading requirements of Section 27.3.5 are now applicable to buildings of all heights.

As discussed in Chapter 31, the wind tunnel procedure should always be considered for buildings with unusual shapes, rectangular buildings with larger aspect ratios, and dynamically sensitive buildings. The effects of torsion can more accurately be determined for these cases and for the more normal building shapes using the wind tunnel procedure.

PART 2: ENCLOSED SIMPLE DIAPHRAGM BUILDINGS WITH $h \leq 160$ ft ($h \leq 48.8$ m)

This section was added to ASCE 7-10 to cover the common practical cases of enclosed simple diaphragm buildings up to height $h = 160$ ft ($h = 48.8$ m). Two classes of buildings are covered by this method. Class 1 buildings have $h \leq 60$ ft ($h \leq 18.3$ m) with plan aspect ratios L/B between 0.2 and 5.0. Cases A through F are described in Appendix D to allow the designer to establish the lines of resistance of the MWFRS in each direction so that the torsional load cases of Fig. 27.3-8 need not be considered. Class 2 buildings have 60 ft $< h \leq 160$ ft (18.3 m $< h \leq 48.8$ m) with plan aspect ratios of L/B between 0.5 and 2.0. Cases A through E of Appendix D are described to allow the designer to establish the lines of resistance of the MWFRS so that the torsional load cases of Fig. 27.3-8 need not be considered.

For the type of buildings covered in this method, the internal building pressure cancels out and need not be considered for the design of the MWFRS. Design net wind pressures for roofs and walls are tabulated directly in Tables 27.5-1 and 27.5-2 using the Directional Procedure as described in Part 1. Guidelines for determining the exterior pressures on windward, leeward, and sidewalls are provided in footnotes to Table 27.5-1.

The requirements in Class 2 buildings for natural building frequency ($75/h$) and structural damping ($\beta = 1.5\%$ critical) are necessary to ensure that the gust-effect factor, G_f , which has been calculated and built into the design procedure, is consistent with the tabulated pressures. The frequency of $75/h$ represents a reasonable lower bound to values found in practice. If calculated frequencies are found to be lower, then consideration should be given to stiffening the building. A structural damping value of 1.5% , applicable at the ultimate wind speeds as defined in the new wind speed maps, is conservative for most common building types and is consistent with a damping value of 1% for the ultimate wind speeds divided by $\sqrt{1.6}$, as contained in the ASCE 7-05 wind speed map. Because Class 1 buildings are limited to $h \leq 60$ ft, ($h \leq 18.3$ m) the building can be assumed to be rigid as defined in the glossary, and the gust-effect factor can be assumed to be 0.85 . For this class of buildings, frequency and damping need not be considered.

C27.5 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

C27.5.1 Wall and Roof Surfaces: Class 1 and 2 Buildings. Wall and roof net pressures are shown in Tables 27.5-1 and 27.5-2 and are calculated using the external pressure coefficients in Fig. 27.3-1. Along-wind net wall pressures are applied to the projected area of the building walls in the direction of the wind, and exterior sidewall pressures are applied to the projected area of the building walls normal to the direction of the wind acting outward, simultaneously with the roof pressures from Table 27.5-2. Distribution of the net wall pressures between

windward and leeward wall surfaces is defined in Note 4 of Table 27.5-1. The magnitude of exterior sidewall pressure is determined from Note 2 of Table 27.5-1. It is to be noted that all tabulated pressures are defined without consideration of internal pressures because internal pressures cancel out when considering the net effect on the MWFRS of simple diaphragm buildings. Where the net wind pressure on any individual wall surface is required, internal pressure must be included as defined in Part 1 of Chapter 27.

The distribution of wall pressures between windward and leeward wall surfaces is useful for the design of floor and roof diaphragm elements such as drag strut collector beams, as well as for MWFRS wall elements. The values defined in Note 4 of Table 27.5-1 are obtained as follows: The external pressure coefficient for all windward walls is $C_p = 0.8$ for all L/B values. The leeward wall C_p value is (-0.5) for L/B values from 0.5 to 1.0 and is (-0.3) for $L/B = 2.0$. Noting that the leeward wall pressure is constant for the full height of the building, the leeward wall pressure can be calculated as a percentage of the p_h value in the table. The percentage is $0.5/(0.8 + 0.5) \times 100 = 38\%$ for $L/B = 0.5$ to 1.0. The percentage is $0.3/(0.8 + 0.3) \times 100 = 27\%$ for $L/B = 2.0$. Interpolation between these two percentages can be used for L/B ratios between 1.0 and 2.0. The windward wall pressure is then calculated as the difference between the total net pressure from the table using the p_h and p_0 values and the constant leeward wall pressure.

Sidewall pressures can be calculated in a similar manner to the windward and leeward wall pressures by taking a percentage of the net wall pressures. The C_p value for sidewalls is (-0.7) . Thus, for $L/B = 0.5$ to 1.0, the percentage is $0.7/(0.8 + 0.5) \times 100 = 54\%$. For $L/B = 2.0$, the percentage is $0.7/(0.8 + 0.3) \times 100 = 64\%$. Note that the sidewall pressures are constant up the full height of the building.

The pressures tabulated for this method are based on simplifying conservative assumptions made to the different pressure coefficient (GC_p) cases tabulated in Fig. 27.3-1, which is the basis for the traditional all heights building procedure (defined as the Directional Procedure in ASCE 7-10) that has been a part of the standard since 1972. The external pressure coefficients, C_p , for roofs have been multiplied by 0.85 , a reasonable gust-effect factor for most common roof framing, and then combined with an internal pressure coefficient for enclosed buildings (± 0.18) to obtain a net pressure coefficient to serve as the basis for pressure calculation. The linear wall pressure diagram has been conceived so that the applied pressures from the table produce the same overturning moment as the more exact pressures from Part 1 of Chapter 27. For determination of the wall pressures tabulated, the actual gust-effect factor has been calculated from Eq. (26.11-10) based on building height, wind speed, exposure, frequency, and the assumed damping value.

C27.5.2 Parapets. The effect of parapet loading on the MWFRS is specified in Section 27.3.5 of Part 1. The net pressure coefficient for the windward parapet is $+1.5$ and for the leeward parapet is -1.0 . The combined effect of both produces a net coefficient of $+2.5$ applied to the windward surface to account for the cumulative effect on the MWFRS in a simple diaphragm building. This pressure coefficient compares to a net pressure coefficient of $1.3G_f$ for the tabulated horizontal wall pressure p_h at the top of the building. Assuming that a lower bound gust-effect factor $G_f = 0.85$, the ratio of the parapet pressure to the wall pressure is $2.5/(0.85 \times 1.3) = 2.25$. Thus, a value of 2.25 is assumed as a reasonable constant to apply to the tabulated wall pressure p_h to account for the additional parapet loading on the MWFRS.

C27.5.3 Roof Overhangs. The effect of vertical wind loading on a windward roof overhang is specified in Section 27.3.4 of Part 1. A positive pressure coefficient of +0.8 is specified. This compares to a net pressure coefficient tabulated for the windward edge Zone 3 of -1.06 (derived from $0.85 \times -1.3 \times 0.8 - 0.18$). The 0.85 factor represents the gust-effect factor G , the 0.8 multiplier accounts for the effective wind area reduction to the 1.3 value of C_p specified in Fig. 27.3-1 of Part 1, and the -0.18 is the internal pressure contribution. The ratio of coefficients is $0.8/1.06 = 0.755$. Thus, a multiplier of 0.75 on the tabulated pressure for Zone 3 in Table 27.5-2 is specified.

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